ChatGPT said:

AI Course-II AIML/REC – Category 3

Practical Lab Tasks – 07/08/2026

Category 3 focuses on **Local Search Algorithms, Local Optima, Global Optima, Exploration and Exploitation**.

I'll provide Python programs in a simple practical-exam format, with aims, algorithms, code, sample outputs and explanations.

Programs covered

No.	Practical task	Python implementation
1	Exploration and Exploitation Trade-offs	Exploration vs Exploitation simulation
2	Hill Climbing – Eight Queens	8 Queens optimization
3	Simulated Annealing – Different Teams	Team formation optimization

1. Analyse Exploration and Exploitation Trade-offs

Aim: To analyse the trade-off between exploration and exploitation in a search space.

Concept:

- **Exploration:** Trying different solutions to discover better possibilities.
- **Exploitation:** Using the best solution found so far to improve the result.

For example, selecting a team:

- Exploration: Try different team combinations.
- Exploitation: Select members from the best-performing combination.

Python Program

Python

```
import random

# Possible options and their rewards
options = [10, 20, 30, 40, 50]

exploration_count = 0
exploitation_count = 0

best_reward = 0

for i in range(20):

    if random.random() < 0.3:
        # Exploration
        choice = random.choice(options)
        exploration_count += 1
        print("Exploration:", choice)

    else:
        # Exploitation
        choice = max(options)
        exploitation_count += 1
        print("Exploitation:", choice)

    reward = choice

    if reward > best_reward:
        best_reward = reward
```

Sample Output

Plain text

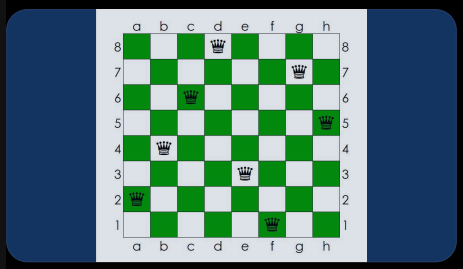
```
Exploration: 20
Exploitation: 50
Exploitation: 50
Exploration: 40
...
Exploration Count: 6
Exploitation Count: 14
Best Reward: 50
```

Note: The output changes because the program uses random selection.

Explanation:

1. Create a list of possible choices.
2. Generate a random number.
3. If the number is below 0.3, perform exploration.
4. Otherwise, perform exploitation.
5. Count both operations.
6. Display the best reward.

2. Hill Climbing Search – Eight Queens Problem



Aim: To implement Hill Climbing Search to solve the Eight Queens problem by placing eight queens on a chessboard so that no two queens attack each other.

Concept:

- A chessboard has 8 rows and 8 columns.
- Place one queen in each row.
- No two queens should share the same column or diagonal.
- The goal is to achieve **zero conflicts**.

Heuristic function:

$$h(n) = \text{Number of attacking queen pairs}$$

Our objective is to minimize the heuristic value.

Python Program

Python

```
import random

N = 8

def heuristic(board):
    conflicts = 0

    for i in range(N):
        for j in range(i + 1, N):

            if board[i] == board[j]:
                conflicts += 1
```

```
            if abs(board[i] - board[j]) == abs(i - j):
                conflicts += 1

    return conflicts


def hill_climbing():

    board = [random.randint(0, 7) for _ in range(N)]

    while True:

        current = heuristic(board)

        if current == 0:
            return board
```

Sample Output

Plain text

```
Eight Queens Solution:

. . Q . . . . .
. . . . . . Q .
. . . . Q . . .
. Q . . . . . .
. . . . . . . Q
. . . Q . . . .
Q . . . . . . .
. . . . . Q . .

Conflicts: 0
```

Explanation:

1. Randomly place eight queens on the board, one per row.
2. Calculate the number of attacking pairs.
3. Move one queen to another column.
4. Calculate the new heuristic value.
5. Select the move that reduces conflicts the most.
6. Repeat until zero conflicts are achieved.
7. If the algorithm reaches a local optimum, restart with a new random board.

Important: Hill Climbing can get stuck in a local optimum. Random restart helps it search for a global solution.

---

3. Simulated Annealing Search – Team Formation

Aim: To implement Simulated Annealing to form two balanced teams based on the skill levels of players.

Use case:

Suppose we have six players with different skill levels. We need to divide them into two teams of three players each.

The objective is to minimize the difference between the total skill levels of both teams.

Concept:

Simulated Annealing is inspired by the cooling process of metals.

- At high temperature, it accepts more changes, including worse solutions.
- At low temperature, it becomes more selective.
- It gradually moves towards an optimal solution.

Python Program

Python

```
team2 = ["D", "E", "F"]


def cost(t1, t2):

    skill1 = sum(players[p] for p in t1)
    skill2 = sum(players[p] for p in t2)

    return abs(skill1 - skill2)


temperature = 100
cooling_rate = 0.95

current_cost = cost(team1, team2)
```

Sample Output

Plain text

```
Team 1: ['A', 'D', 'F']
Team 2: ['B', 'C', 'E']

Team 1 Skill: 190
Team 2 Skill: 200

Skill Difference: 10
```

The exact output may vary because of random selection.

Explanation:

1. Create six players with different skill levels.
2. Divide them into two teams.
3. Calculate the difference between their total skills.
4. Randomly select one player from each team.
5. Swap the selected players.
6. Calculate the new difference.
7. Accept better solutions.
8. Sometimes accept worse solutions based on temperature.
9. Reduce the temperature gradually.
10. Display the final teams.

Important concepts for practical viva

Concept	Meaning
Local Search	Searches for a better solution using neighboring states.
Global Search	Explores the wider search space to find a global solution.
Local Optimum	A solution better than its immediate neighbors, but not necessarily the best overall.
Global Optimum	The best possible solution in the entire search space.

Exploration	Trying new solutions.
Exploitation	Improving or using the best-known solution.
Hill Climbing	Moves towards a better neighboring solution.
Simulated Annealing	Sometimes accepts worse solutions to escape local optima.
Heuristic Function	Estimates the quality of a solution.
Temperature	Controls the probability of accepting worse solutions in Simulated Annealing.

Important viva questions

1. **What is Hill Climbing?**
   A local search algorithm that continuously moves towards a better neighboring state.
2. **What is a local optimum?**
   A solution that is better than its neighboring solutions but may not be the best overall.
3. **What is a global optimum?**
   The best solution in the entire search space.
4. **What is the main disadvantage of Hill Climbing?**
   It can get stuck in local optima.
5. **How can we overcome local optima?**
   By using random restart or Simulated Annealing.
6. **What is Simulated Annealing?**
   An optimization algorithm inspired by the cooling process of metals.
7. **Why is temperature important?**
   It controls the acceptance of worse solutions.
8. **What happens when temperature decreases?**
   The probability of accepting worse solutions decreases.
9. **What is exploration?**
   Searching for new possible solutions.
10. **What is exploitation?**
    Improving the best-known solution.
11. **What is the heuristic function in Eight Queens?**
    The number of pairs of attacking queens.
12. **What is the goal of Eight Queens?**
    Place eight queens without any two queens attacking each other.

Comparison of the algorithms

Feature	Hill Climbing	Simulated Annealing
Search type	Local search	Local search
Accepts better solution	Yes	Yes
Accepts worse solution	No	Sometimes
Escapes local optimum	Difficult	Possible
Uses temperature	No	Yes
Main objective	Improve neighboring state	Find a near-global or global optimum

My recommendation for your lab exam:

Learn these three programs in this order:

1. Exploration and Exploitation – easiest.
2. Eight Queens using Hill Climbing – important for understanding heuristics.
3. Team Formation using Simulated Annealing – important for understanding temperature and probability.

Remember: Category 3 is mainly about understanding how search algorithms find better solutions and how they avoid getting trapped in local optima.